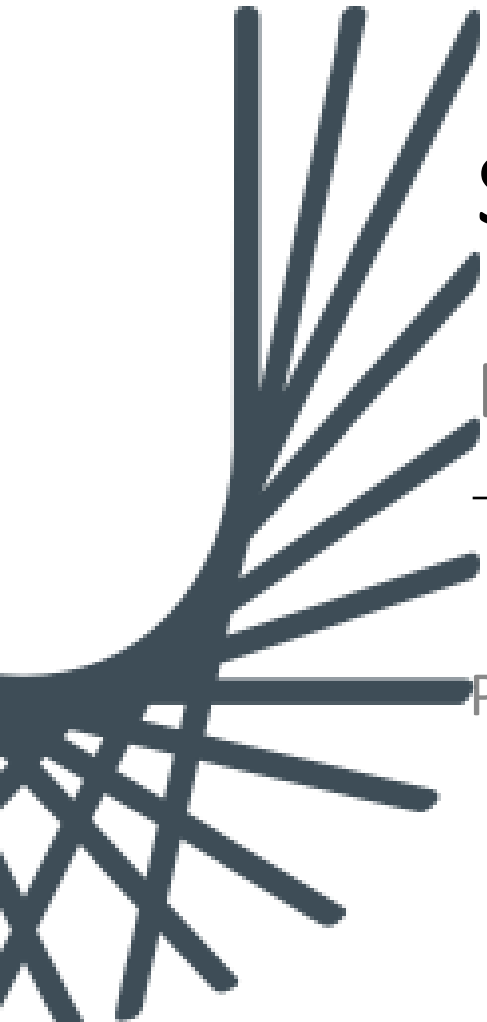




Searching

Basic searching algorithms

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Main concepts to be covered in this section

- Searching
 - Linear Search (unsorted list)
 - Binary Search (sorted list)

Linear and binary searching



Searching

- Generally a search method will look something like (ignore access modifier):

```
int search(int[] a, int size, int lookingFor)
```

- This method would return the position in the array of the value 'lookingFor' if it exists and -1 otherwise.

Searching

- Before we decide on how to search through a list we need to know whether or not the list is sorted.
 - Not Sorted → Linear Search
 - Sorted → Binary Search

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- Searching
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Linear Search

```
public int linearsearch(int[] intArray, int size, int lookingFor)
{
    int i = 0;

    while (i < size)
    {
        if (intArray[i] == lookingFor)
            return i; //this terminates the method
        i++;
    }
    return -1; // we only get to here if we have got to
               // end of list and never found item
}
```

Main concepts to be covered in this section

- Searching
 - Linear Search (unsorted list)
 - Binary Search (sorted list)

Binary Search

- If the list is sorted we make use of this (same way we search through telephone book)
- The binary search is such a search and is a particularly efficient search.

Binary Search Algorithm

- The overall algorithm works by always looking at the midpoint of the list.
- If we have found what we are looking for, great.
- If not...

Binary Search Algorithm contd.

- If not...
- ...and the value we are looking for is less than the midpoint value...
- ...then, because the list is sorted, we can 'throwaway' all the values greater than the midpoint. So we 'redraw' our list to be the left-hand part of the list and repeat the procedure.

Binary Search Algorithm contd..

- If not...
- ...and the value we are looking for is greater than the midpoint value...
- ...then (using the same logic) we 'redraw' the list to just include the right-hand part of the list and repeat the procedure.

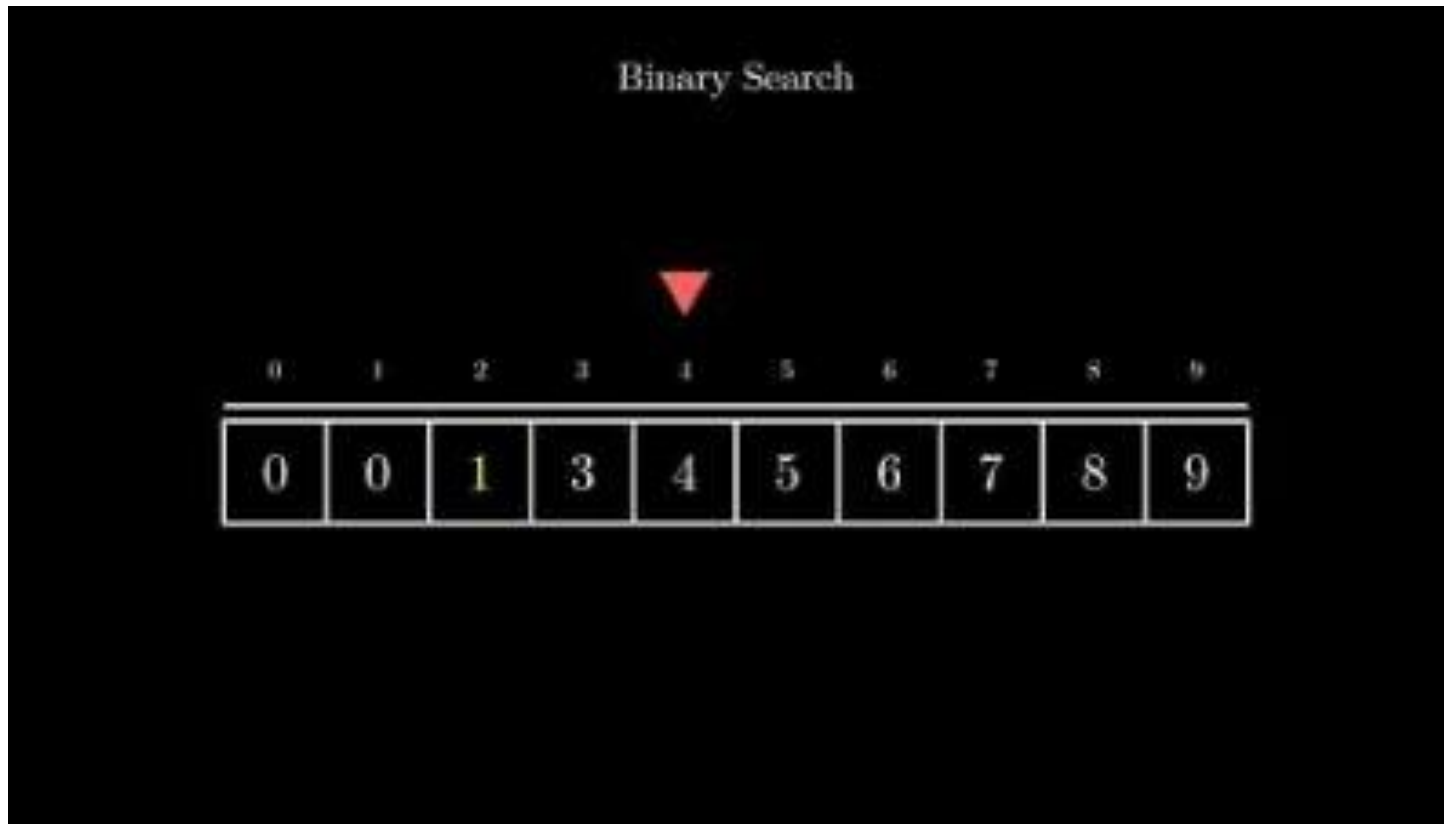
Binary Search Code

```
public int binarysearch(int[] intArray, int size, int lookingFor)
{

    int mid;
    int left = 0;
    int right = size-1;

    while (left <= right)
    {
        mid = (left + right) / 2;
        if (intArray[mid] == lookingFor)
            return mid; //this terminates the method
        else if (intArray[mid] > lookingFor) // element in left hand of list, if it exists.
            right = mid - 1;
        else if (intArray[mid] < lookingFor) // element in right hand of list, if it exists.
            left = mid + 1;
        }
        return -1; //this terminates the method because we have gone through all list and it's not there
    }
```

Animation of both searches



Problems
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**Any
Questions?**



